



Drug Repurposing with an Intelligent Orchestrating Platform

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INTRODUCTION

Drug repurposing demands the simultaneous integration of transcriptomic, structural, and knowledge-based evidence — a challenge beyond the reach of isolated AI tools. We therefore developed an AI-orchestrating, multimodal platform that unifies gene expression analysis, protein structure modeling, knowledge graph reasoning, and LLM synthesis to systematically identify and prioritize repurposing candidates across diverse disease contexts.

Rather than relying on any single modality, the platform coordinates multiple specialized AI tools, each customized and optimized for drug repurposing, within a coherent orchestration framework, enabling each data stream to contribute complementary evidence toward candidate prioritization. The modular design ensures the platform can continuously incorporate advances in AI modeling, improving predictive capacity as the field evolves.

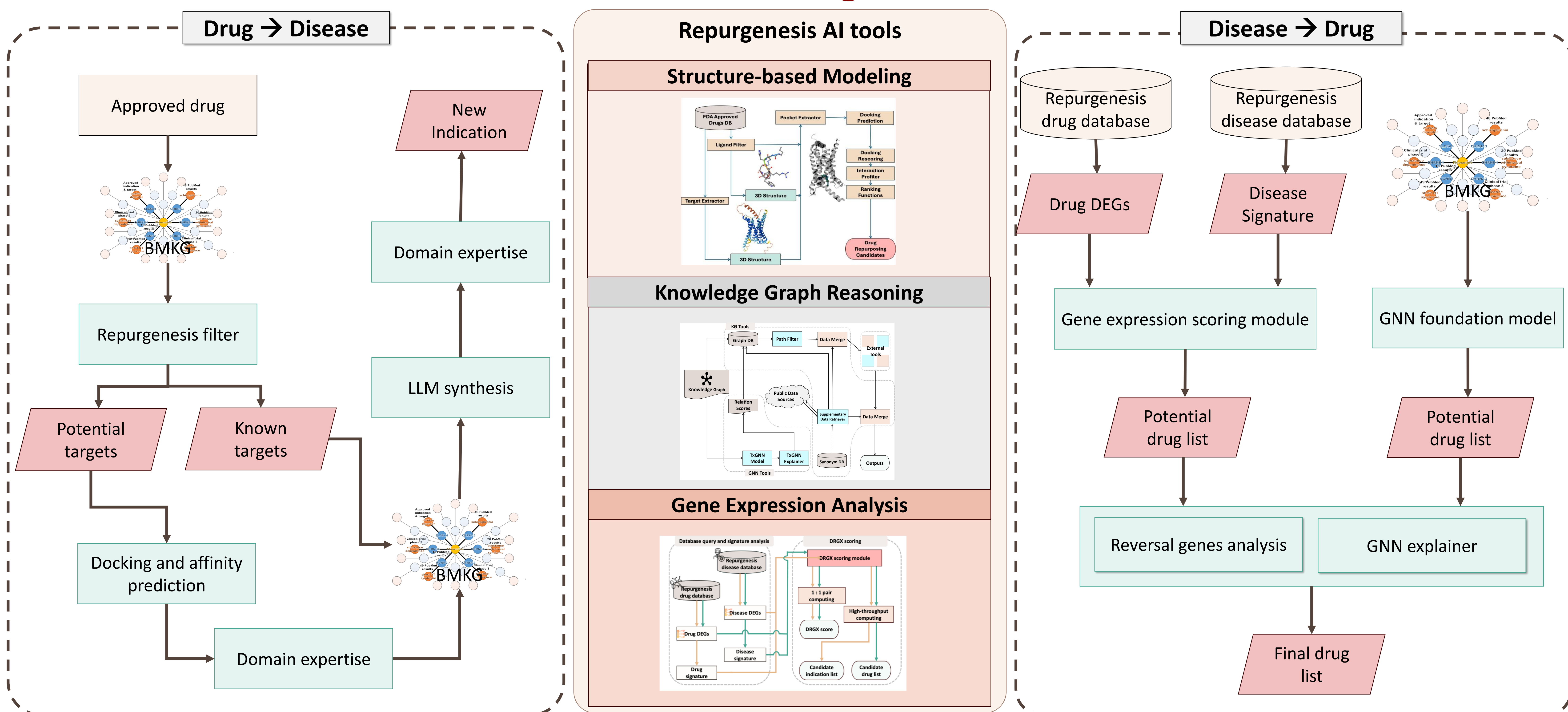
Early validation demonstrates that these orchestrating AI tools yields high-confidence repurposing candidates, with computational predictions independently confirmed through experimental assays — establishing a reproducible path from AI-driven discovery to translational evidence.

KEY HIGHLIGHTS

Platform strengths The platform employs multiple specialized AI tools, each customized and optimized for drug repurposing, coordinated within a unified multimodal framework. An approved therapeutic candidate was identified for vitiligo and confirmed through in vitro validation, and is currently under patent filing.

Dual workflows *Drug-to-Disease*: Initiates from an approved drug; maps drug-target paths via BMKG to identify targets, then evaluates candidates through protein structure, and domain expertise to surface novel indications. *Disease-to-Drug*: Begins from disease and drug signatures; applies DRGX scoring and a GNN foundation model to generate candidate drug lists, further refined through reversal gene analysis and GNN-based explainability.

AI-Orchestrating Platform



RESULTS

Translational Pipeline (From Discovery to IP):

Disease Area	Indication	Drug ID	PoC Validation	IP Filing	Pre-Clinical
Autoimmune	Vitiligo				
Fibrosis	Idiopathic Pulmonary Fibrosis				
Neurology	Spinocerebellar Ataxia				

Leveraging existing safety profiles, our pipeline accelerates repurposing from AI discovery to clinical and IP outcomes.

Case Study in Drug β :

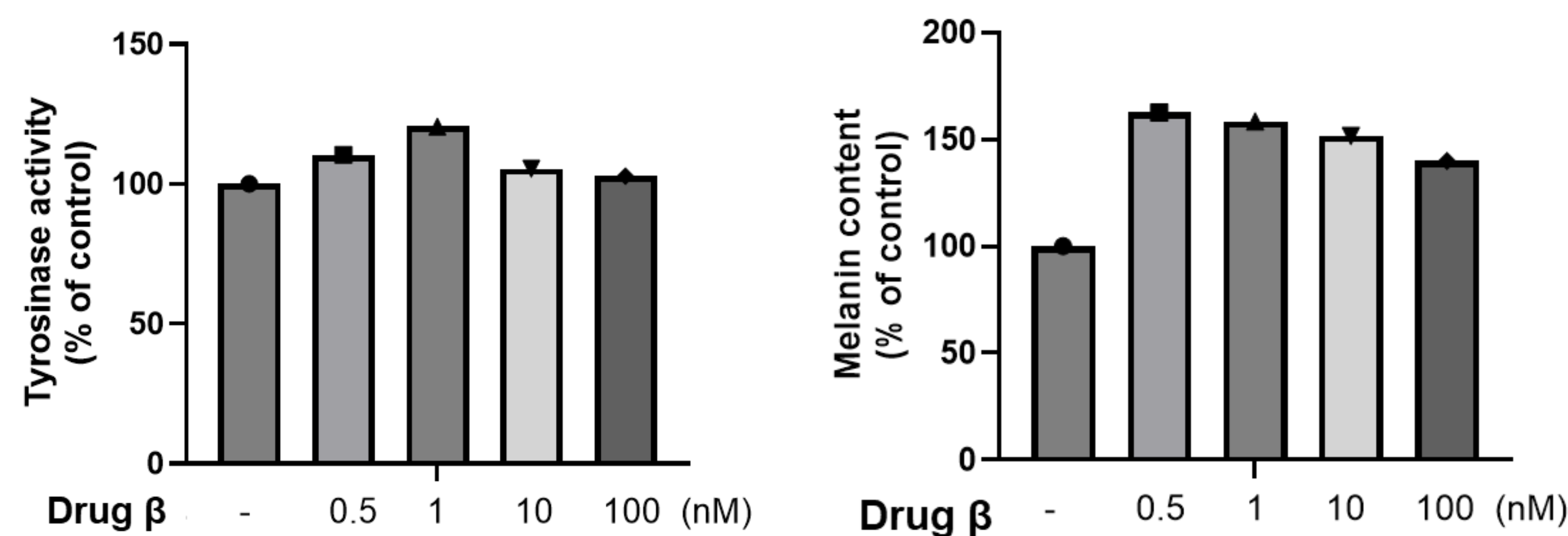
A currently approved therapy, originally indicated for a non-dermatological condition, was identified by the AI-orchestrating platform as a candidate for vitiligo — an autoimmune disease characterized by skin depigmentation — using the Drug-to-Disease workflow. Drug β increased tyrosinase activity and cellular melanin content in B16F10 cells, providing direct functional evidence for melanogenic potential. This case study demonstrates that the AI-driven repurposing candidate was independently confirmed through both computational modeling and in vitro experimentation, validating the platform's translational potential.

Binding Prediction Result in Drug β :

Affinity Prediction					
Drug	Target	confidence	pIC50	logIC50	Binding Prob.
Drug β	GPCR_1	0.845939	11.404851	-2.361328	0.687734
	4PM7_1	0.844177	9.019709	-0.612690	0.615548
	3VHU_1	0.708079	8.671108	-0.357117	0.511958
	4NTX_1	0.656728	8.888714	-0.516653	0.632282
Drug A	GPCR_1	0.820489	6.475383	1.252651	0.280430
	4PM7_1	0.941027	5.515185	1.956609	0.377265
	3VHU_1	0.815950	5.861539	1.702684	0.287462
	4NTX_1	0.641424	6.737264	1.060657	0.343843

Target Docking and Pose Prediction				
Target	Drug	Pose Rank	Confidence	Best Pose
GPCR_1	Drug β	Pose_rank01	-1.649114	
		Pose_rank02	-1.971637	
		Pose_rank03	-2.086164	
		Pose_rank10	-3.214103	
GPCR_1	Drug A	Pose_rank01	0.370985	
		Pose_rank02	0.349616	
		Pose_rank03	0.322107	
		Pose_rank10	-0.407854	
GPCR_1	Drug C	Pose_rank01	-0.538920	
		Pose_rank02	-0.538920	
		Pose_rank03	-0.538920	
		Pose_rank10	-0.538920	

Drug β in Cell Model:



- Drug β increased tyrosinase activity in B16F10 cells.
- Drug β increased cellular melanin content in B16F10 cells.

REFERENCE

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